

Sessional	I
Program	B. Tech.
Class	4 th Semester
Subject	Strength of Materials
Code	PIPC 211
Max Marks	15
Date/Time	21.02.2026/11:00 AM - 11:50 AM

Note: Attempt all the questions. Assume suitably and state, additional data required, if any

1. A steel rod, 12 mm in diameter centrally passes through a copper tube 2.5 m long and having 36 mm and 48 mm as internal and external diameters respectively. The tube is closed at each end by 24 mm thick steel plates which are secured by nuts. The nuts are tightened until the copper tube is reduced in length by 0.50 mm. The whole assembly is then raised in temperature by 60°C . Calculate the stresses in copper and steel before and after the rise of temperature, assuming that the thickness of plates remains unchanged.

Take $E_s = 210 \text{ GN/m}^2$, $E_c = 105 \text{ GN/m}^2$,

$\alpha_s = 1.2 \times 10^{-5} \text{ per } ^{\circ}\text{C}$, $\alpha_c = 1.75 \times 10^{-5} \text{ per } ^{\circ}\text{C}$

(7)

2.

- a. State different theories of failure and give their graphical representation. (2)

- b. At a point in stressed material the direct stresses on two perpendicular planes are 140 MN/m^2 compression and 90 MN/m^2 tension respectively, and the shearing stress on these planes is τ . The yield stress for the material is 950 MN/m^2 . Find the value of τ at which failure may be expected according to different theories of failure. Take factor of safety = 5 and Poisson's ratio as 0.3. (6)